

Do biased perceptions of wealth distribution affect savings behavior?

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Abstract

We estimate the relationship between people’s biased perceptions of their rank in the wealth distribution and savings behavior. Using survey data from Austria, we show that there is a relationship between misperceptions and savings that is statistically significant at least at the 5% level. A one-decile downward deviation in perceived rank, relative to one’s decile based on observed wealth, is associated with a 5% increase in the savings rate evaluated at the mean (predictive estimate). When using an instrumental-variables approach, the estimated effect increases substantially to 18%. Preferences that feature relative wealth in the utility function can explain this relationship. Our findings inform contemporary macroeconomic models and have important implications for policy.

JEL Codes: D12, D31, E21, J62

Keywords: Subjective wealth perception, Wealth distribution, Savings rate, Economic decision-making, Instrumental variable (IV) approach

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1 Introduction

The role of the distribution of income and wealth in determining aggregate savings receives considerable attention against the backdrop of the decline in the natural interest rate. While prior studies focus on inequality in actual income, we highlight the significance of individuals’ subjective perceptions of distributional outcomes for economic decision-making. Capitalizing on a survey instrument where individuals rank themselves in the national distribution of wealth, we show a strong relationship between perceived relative wealth ranks and savings. Our findings are at the core of economics, the saving decision in consumer demand. Therefore, the study informs contemporary macroeconomic models (predictive estimate) and helps to evaluate the impact of social segregation and information bubbles on economic decisions influenced by individual perceptions of social status or any other phenomenon that changes the perception of relative social status (causal effect).

Findings Our main finding is based on the observation that individuals have substantial misperceptions of their rank in the wealth distribution, and consists of a key novel empirical insight: individuals who underestimate their rank in the wealth distribution (“underestimators”) have systematically higher savings rates compared to their peers with accurate perceptions or an upward bias in their perceived wealth rank (“overestimators”). This finding suggests that misperceptions in wealth ranks, that have begun to attract scholarly interest recently, are not merely an artifact of the survey setup and social desirability in the response behavior, but that economic behavior changes with biased perceptions.

Across a broad range of specifications, we find that perception bias, defined as perceived wealth decile minus actual wealth decile, is negatively associated with savings. Our simplest predictive estimate indicates that a one-decile change in bias toward underestimation is associated with a 5% increase in the savings rate (2.07 percentage points). This effect is more pronounced for negative bias: Our graphical evidence shows that underestimating one’s wealth rank (irrespective of the magnitude of the bias) goes along with an 11 percentage point higher savings rate (26% of the savings rate). We also employ a novel instrumental variable (IV) approach and leverage a wage transparency policy which leads individuals to perceive themselves higher up in the wealth ladder to address common support and other challenges to identification.¹ Using this strategy,

¹As the common support of underestimators and overestimators with regard to the wealth rank is sparse, our regression results that establish predictive effects rest on strong extrapolation outside the common support.

we establish that the savings rate increases by 7.7 percentage points (18% of average savings) for one decile of less favorable self-perception in the wealth distribution, which confirms and strengthens our original result. The IV approach is less sensitive to the common support issue as it only rests on the differences in perceived rank explainable by the wage transparency law. It also suggests that our finding is driven neither by reverse causality nor by unobserved heterogeneity.

The main predictive and causal estimates are not only precisely estimated (with p-values at least at the 5% level), but also economically meaningful. For example, the 2.07 percentage points savings premium (predictive estimate) for a one-decile downward deviation of one's perceived wealth rank at a given actual decile implies a substantial reduction in consumer spending in favor of saving. With a mean deviation of -1.38 in 2017 and an average savings rate of 42% (1357.40) before spending on consumer durables, this implies an additional 92.26 monthly. According to our causal estimate (7.7 percentage points), correcting misperceptions in a household with an average level of bias (i.e. underestimation by 1.38 deciles) would increase its monthly savings by 343.43.

Heterogeneity in savings rates The robust evidence on the relationship between misperceptions and savings empirically established in this paper for the first time informs recent debates in macroeconomics on the role of heterogeneity for aggregate economic outcomes. As the natural rate of interest (r^*) falls, trends such as the high savings rates at the top of changing income and wealth distributions have pivoted to the center of the debate (Mian, Straub, and Sufi 2021; Summers 2015). Many current empirical studies document substantial heterogeneity in consumption out of additional income (Lewis, Melcangi, and Pilossoph 2024; Crawley and Kuchler 2023; Fagereng, Holm, and Natvik 2021; Jappelli and Pistaferri 2020; Christelis et al. 2019; Jappelli and Padula 2016). One of the most robust findings on propensities to consume and heterogeneity is the strong association between income or cash-on-hand (sum of current income and wealth) and savings choices (Boehm, Fize, and Jaravel 2025; Garbinti, Lamarche, and Savignac 2024; Gelman 2021; Fisher et al. 2020; Jappelli and Pistaferri 2014; Misra and Surico 2014). Aguiar, Bills, and Boar (2024) argue that preference heterogeneity can explain differences in savings propensities along the wealth distribution. In addition to these contributions, this paper shows that savings behavior is driven by perceptions even when controlling for income and wealth.² As perceptions are correlated with both, we offer evidence that

²By stressing the important role of perceptions of outcomes in addition to actual outcomes, our work is also an extension of an important emerging literature that considers the measurement of individuals'

a part of the variation previously attributed to income and wealth heterogeneity may result from perceptions about relative wealth.

Misperceptions Recently, individuals’ biased perceptions of their relative affluence and the consequences of biased beliefs have drawn substantial attention. The key finding is that individuals across the income distribution have biased perceptions. Most studies find respondents to believe that their income is in the center of the distribution (Hvidberg, Kreiner, and Stantcheva 2023; Hoy and Mager 2021; Cruces, Perez-Truglia, and Tetaz 2013) (compression effects, centre-bias or middle-class bias), though asymmetric misperceptions where most people underestimate their rank in the income distribution have also been documented (Karadja, Mollerstrom, and Seim 2017).³ Regarding wealth, three studies are closely related to our paper.⁴ First, Douenne, Sund, and Weele (2024) elicit misperceptions of income and wealth ranks in a survey-experiment. Even though they are able to connect income and wealth perceptions to administrative data on actual income and wealth, their results parallel our finding that there is a strong compression in the wealth rank perceptions.⁵ The authors argue that misperceptions and information treatments have little effect on policy preferences. This contrasts with the findings of a second related paper (Albacete, Fessler, and Lindner 2022), showing that the effects of information treatments depend on the direction of bias in wealth perceptions. Third, there is evidence that wealth rank perceptions affect people’s willingness to take risk (Fehr and Reichlin 2023). Our findings do not only reinforce previous research showing a strong bias towards the middle with a majority of underestimators when individuals estimate their rank in the wealth distribution, but we also connect this phenomenon to

beliefs and other behavioral factors in order to understand macroeconomic phenomena. Bounded rationality affects the predictions of standard macroeconomic models in several ways (Farhi and Werning 2019; Hommes 2021): Not least, myopic individuals consume more out of a government transfer than rational Ricardian agents (Gabaix 2020). Moreover, Andre et al. (2022) show that people’s subjective models of the macro-economy drive forecasts causing disagreement in expectations. There is also evidence that individual attention is a key determinant of expectations (Link et al. 2023).

³Windsteiger (2022) shows how residential segregation contributes to income rank perceptions clustering around the middle of the distribution, feeding into policy preferences.

⁴Complementing evidence on perceived wealth ranks, several papers elicit survey respondents’ knowledge about wealth inequality. Evidence for the US suggests that individuals underestimate the share of wealth held by the wealthiest quintile by approximately 25 percentage points (Norton and Ariely 2011), while Dutch respondents underestimate the share of wealth held by the top decile by 12.2 percentage points. In contrast, Stantcheva (2021) finds that respondents slightly overestimate the share of wealth held by the wealthiest percentile of the US wealth distribution, by approximately seven percentage points. This work differs from the approach of this paper, which focuses on perceived ranks in the wealth distribution. An individual may have an accurate perception of their rank in the wealth distribution, while still under- or overestimating the share of wealth held at the top and vice versa.

⁵Note that there are more underestimators than overestimators in their sample, echoing our findings.

a new response margin: savings. This sheds new light on the nature of misperceptions. For example, Douenne, Sund, and Weele (2024) argue that the compression they observe may be due to cognitive uncertainty (Enke and Graeber 2023), where individuals regress to an intermediate option when they face uncertainty. The fact that misperceptions continue to predict saving even after conditioning on actual wealth suggests that misperceptions of wealth ranks may not purely result from individuals struggling with a difficult survey question.

Social comparisons and behavior Our empirical contributions also matter for understanding the implications of social comparisons more generally for economic behavior. Previous work that is closely related to this paper assumes a link between relative wealth and savings behavior (Michaillat and Saez 2021; Corneo and Jeanne 1997), while we provide direct evidence on this mechanism. We show that our findings are consistent with a theoretical model where relative wealth is an argument in the utility function, which is also a key element in the theory of optimal capital taxation advanced by Saez and Stantcheva (2018). A growing empirical literature that does not consider wealth specifically studies how social comparisons relate to consumption and financial decisions (D’Acunto, Rossi, and Michael Weber 2024; Schwerter 2024; Bricker, Krimmel, and Ramcharan 2021; Agarwal, Qian, and Zou 2021; Agarwal, Mikhed, and Scholnick 2020; Georgarakos, Haliassos, and Pasini 2014) as well as political choices (Fehr, Mollerstrom, and Perez-Truglia 2022; Fisman, Kuziemko, and Vannutelli 2021; Roth and Wohlfart 2018; Hauser and Norton 2017; Cruces, Perez-Truglia, and Tetaz 2013; Clark and Senik 2010). This literature has not considered private wealth as a dimension of ranking yet.

Contribution In this paper, we break new ground by providing evidence on the consequences of wealth rank misperceptions. We measure bias in wealth perceptions and offer an analysis of the consequences of such biased perceptions for savings behavior. By providing our predictive effects we deliver a characterization of (conditional) differences between savings rates by perception of relative wealth rank which can be used to inform macroeconomic models. The robustness of the effect to different model specifications, savings definitions and the novel instrumental variable identification strategy suggests that the paper provides a tool to evaluate the effects of policy interventions, developments or events which have an impact on the savings rate that is mediated through the perception of relative wealth.

Policy The findings open up interesting questions for policy design. For example, it may be possible to employ targeted information treatment tools to manage consumption from a macroeconomic perspective. Moreover, as subjective assessments of people’s own position in the wealth distribution account for high savings rates in certain parts of the population, changes in the determinants of misconceptions could change the transmission mechanism of monetary policy. One may think of similar interventions in the area of financial stability and strengthening household resilience. At the same time, the rise of social media and AI may provide an even stronger amplifier of differences in perceptions and resulting differences in economic behavior. Our causal estimates allow to quantify the effects on savings for any change in relative wealth rank perceptions.

Roadmap The remainder of the paper is organized as follows. In Section 2, we develop a minimal model of savings decisions that incorporates perceived wealth ranks. Section 3 outlines our identification strategy, while Section 4 introduces the data, presents descriptive statistics, and documents the extent of bias in respondents’ perceived wealth ranks. The main empirical results are reported in Section 5, where we also provide extensive robustness checks and additional analyses. We conclude in Section 6.

2 A minimal model

We set up a minimal two-period model to illustrate a potential mechanism introducing the subjective wealth distribution into a standard framework.

Setup There are two periods. Individuals earn income Y in period 1 (working age), consume C in period 1, and save $S = Y - C$ for period 2 (retirement). S represents consumption in period 2 and constitutes wealth before retirement. Let $F(\cdot)$ be individuals’ subjective expectation of the distribution function of wealth across the population. $F(S)$ is therefore the individuals’ subjective expected rank of their own level of wealth S in the wealth distribution F . Restricting the model further, we assume that individual beliefs are of the form $F(s) = F(s, \Delta) = \min(\max(F_0(s) + \Delta, 0), 1)$, where F_0 is the true distribution of wealth, and Δ is a shift in perceived ranks, capturing mistaken beliefs.

A common explanation for pervasive misperceptions is that agents may rely on heuristics (Thaler 2016) when they draw inferences about their relative wealth. Relative status comparisons are prone to the representativeness heuristic, arising from individual failure of applying Bayes’ rule to information that people receive (Kahneman and Tversky 1972).

Preferences Individuals have a preference over consumption in either period. They might additionally have a preference over wealth ranks (Michaillat and Saez 2021; Corneo and Jeanne 1997; Bakshi and Chen 1996). Individuals choose savings S to maximize utility

$$U(S, \Delta) = u(C) + u(S) + v(F(S, \Delta)) = u(Y - S) + u(S) + v(F(S, \Delta)),$$

where in the first two terms, $u(\cdot)$, reflect the utility from consumption, assumed to be strictly increasing and concave ($u'(C) > 0$, $u''(C) < 0$). The third term, $v(\cdot)$, is non-standard, denotes a preference and represents the utility from perceived wealth ranks, concave over $[0, 1]$.⁶ Our main theoretical result below will characterize the dependence of savings on belief distortions. For such a dependence to exist, we require a decreasing marginal utility of perceived wealth ranks, in addition to the presence of belief distortions.

Utility maximization, given the assumed form of belief distortions, implies the first order condition $\partial U(S, \Delta) / \partial S = 0$, which can be written as

$$u'(Y - S) = u'(S) + v'(F_0(S) + \Delta) \cdot F'_0(S),$$

assuming that the utility maximizing S is such that $0 < F(S, \Delta) < 1$.

This condition states that the marginal utility of first period consumption (from income not saved) equals the sum of the marginal utility of second period consumption, plus the product of the marginal utility of perceived wealth ranks and the derivative of the true distribution of wealth. It balances the marginal utility of consumption today against the marginal utility of consumption in the future (savings today) plus the utility derived from perceived wealth rank.

To derive comparative statics of savings S with respect to belief distortions Δ , consider the derivative of the first order condition with respect to Δ . We get

$$\frac{\partial^2 U(S, \Delta)}{\partial S^2} \cdot \frac{\partial S}{\partial \Delta} + \frac{\partial^2 U(S, \Delta)}{\partial S \partial \Delta} = 0,$$

⁶Preferences for wealth have become increasingly popular in the recent writing in economics. Saez and Stantcheva (2018) review different microfoundations for introducing wealth in the utility function: Bequest motives, entrepreneurship and service flows from wealth. They also stress that preferences for wealth feature in the history of economic thought. Max Weber (1958) and Keynes (1931) argue that individuals value wealth as such. Smith (1982) suggests that the pursuit of social status incentivize people to accumulate wealth. This paper builds on these arguments. However, in our model, it is the perceived rank that matters for utility, rather than the true rank in the distribution of wealth.

where $\frac{\partial^2 U(S, \Delta)}{\partial S^2} < 0$ by the second order condition for optimal S , and

$$\frac{\partial^2 U(S, \Delta)}{\partial S \partial \Delta} = v''(F_0(S) + \Delta) \cdot F'_0(S),$$

where in turn $F'_0(S) \geq 0$, and v'' is negative by assumption on the decreasing marginal utility of ranks. The derivative of $u'(S)$ with respect to Δ is zero, by construction.

Combining these equalities and inequalities, we get

$$\frac{\partial S}{\partial \Delta} = -\frac{v''(F_0(S) + \Delta) \cdot F'_0(S)}{\frac{\partial^2 U(S, \Delta)}{\partial S^2}} \leq 0 \quad (1)$$

Based on this setup, we derive observational implications that underscore the impact of subjective wealth perceptions on savings behavior, bridging traditional economic assumptions with insights from behavioral economics. The features of our minimal model directly relate to belief distortions possibly based on biased extrapolation from a segregated social environment and utility from relative rank or status.

Observational implications The framework introduced leads to the following two observational implications. One based on the standard model and one based on the behavioral extension. Transitioning from the classical to a behavioral framework by incorporating Δ reveals contrasting outcomes. While classical assumptions predict uniform savings behavior irrespective of subjective perceptions, the inclusion of $v(\cdot)$ and Δ introduces variability in savings decisions, underscoring the influence of individuals' beliefs about their wealth rank:

1. For the **neoclassical reference model**, $v \equiv 0$, and therefore the first order condition becomes

$$u'(Y - S) = u'(S).$$

In particular:

- (a) Savings increase in income, but less than proportionally, if u is increasing and concave.
 - (b) The distortion in perceived wealth ranks Δ does not affect savings.
2. For a model with preferences over wealth ranks, we have $v \neq 0$. As v is concave in the perceived rank, $v'(F_0(S) + \Delta)$ is a decreasing function in Δ . We get the following comparative statics of savings decisions:

- (a) As in the neoclassical model savings increase in income.
- (b) In contrast to the neo-classical model, savings are decreasing in Δ . To see this, consider Equation 1. The formula demonstrates that an decrease in Δ (believing one is worse off than they are) leads to greater savings.

Given this minimal theoretical framework, we hypothesize that (i) in accordance with standard economic theory, savings will increase with income and wealth (permanent income) and (ii) particularly, and in addition to standard economic theory, underestimating one's position in the wealth distribution will be associated with an increased savings rate. Hypothesis (ii) translates to empirically testing the null hypothesis $\frac{\partial S}{\partial \Delta} = 0$ as in the standard model against the alternative $\frac{\partial S}{\partial \Delta} < 0$.

3 Identification

The key goal of our analysis is to recover an estimate of the elasticity $\frac{\partial S}{\partial \Delta}$. Empirically, this corresponds to the following naive model for individual respondent i with savings rate S_i and individual and household level characteristics X including income and wealth that may be related to Δ and also affect S_i (and η_i the error term):

$$S_i = \alpha_0 + \alpha_1 \Delta_i + X\beta + \eta_i \quad (2)$$

We present graphical evidence and OLS estimates of this relationship, and appraise robustness by varying the variables in X , for example. However, there are several likely threats to the identification of α_1 .

One of the methodological challenges we encounter in our analysis is the issue of common support for over- and underestimators of wealth rank. Naturally there are no overestimators at the very top and no underestimators at the very bottom of the distribution.⁷ Beyond that, large areas with sparse support that contain predominantly either over- or underestimators exist (Tables 11, 12, and 13 in the Online Appendix A). As a result, it is difficult to draw reliable inferences about the effect of being an over- or underestimator at a particular point of the wealth distribution. As we lack full common support, we are essentially forced to extrapolate the effect outside the range of the data, which can introduce bias and reduce the reliability of our estimates. In this context our IV approach becomes particularly useful. It is less sensitive to the common support

⁷In the Online Appendix E, we show that dropping the first and the tenth decile from the analysis yields coefficients that are smaller in magnitude but still highly statistically significant.

issue as it only rests on the differences in perceived rank explainable by the instrument that lie with the common support.

In Equation 2, several potential sources of endogeneity arise. On the one hand, unobserved heterogeneity may be a factor that drives a simple correlation between savings rates and misperceptions. For example, it is conceivable that expectations, including future inheritance receipts, are the reason why some individuals have distorted beliefs about their wealth rank (because they may rely on their wealthy parents as a reference group) but at the same time feel little need to save. Notably, such factors would tend to bias the relationship in the opposite direction to our hypothesized positive effect of underestimation on savings. Another source of unobserved heterogeneity may be the cost of information acquisition. In a world with imperfect information, it can be costly to acquire information. As a result, some agents will have biased perceptions about inequality and their position in the distribution (Cruces, Perez-Truglia, and Tetaz 2013). If the individual cost of acquiring additional information correlates with both misperceptions and savings behavior, our estimates are biased. Endogeneity may also arise from the fact that individuals may take their ability to save as an indicator of their relative affluence. In this case, one would expect causation to go from savings to misperceptions, rather than in the other direction. Employing an IV approach addresses this concern. With a valid instrument, the variation used to estimate α_1 in Equation 2 is orthogonal to omitted factors and the respondent's savings.

Our IV approach builds on a pay transparency law introduced to combat the gender pay gap, implemented in Austria between 2010 and 2014. The law mandates firms to disclose the pay structure by gender to employees (within the company).⁸ The key idea of our IV approach is that the policy induced variation in perceptions of relative affluence between 2010 and 2014 that can be used for identification. As the transparency policy only affects large firms (more than 150 employees), we expect employed individuals who live in areas with more large firms to be better informed about their relative affluence than individuals in other areas after the policy implementation, but not before. Therefore, our instrumental variable is the interaction of the share of large firms in a district with a post-reform dummy variable (conditional on year fixed-effects and district

⁸The law requires biannual reporting of mean or median earnings, broken down by job type and gender. Earnings includes all income components, including overtime pay, in-kind payments, bonus payments and incentive payments. The job type classification is based on the typologies in the collective bargaining agreements which cover more than 90% of all employees. The job classification distinguishes between different skill levels and -types. For data protection reasons, one earnings group must include at least three employees for reporting. Pay reports are internal company information, and subject to confidentiality rules. An example and further details of the anonymized pay report are available online: <http://web.oegb.at/S06/frauen/DerEinkommensbereichEinPraxisratgeber.pdf>.

fixed-effects). Using the interaction between the timing of a reform and its heterogeneous effects across geographical units makes our instrumental variable related to several other papers that exploit similar designs (De Cao and La Mattina 2019; Larreguy and Marshall 2017; Enikolopov, Petrova, and Zhuravskaya 2011). Crucially, our approach does not rely on the share of large firms for exogenous variation, which is unlikely to hold (Fagereng, Guiso, and Pistaferri 2017).

The first stage of this IV approach rests on two empirical facts for instrument relevance. First, the pay transparency law must affect individual perceptions of relative affluence. Second, we require that individuals extrapolate from their position in the distribution of salaries to their wealth rank, because the pay transparency law is about salaries. Regarding the former, an evaluation of the Austrian pay-transparency policy’s labor market effects based on administrative data and an event-study design suggests that the policy did not affect pay, but reduced job separations (Gulyas, Seitz, and Sinha 2023). As quits are strongly associated with employee perceptions about the fairness of pay schedules (Dube, Giuliano, and Leonard 2019), the fall in job separation is likely to result from employees facing a better relative pay situation than they anticipated (Gulyas, Seitz, and Sinha 2023).⁹ Therefore, individuals may update their perceptions of relative affluence. Regarding extrapolation from salaries to wealth, there is strong evidence that individuals infer their position in the wealth distribution based on their income rank. For example, Douenne, Sund, and Weele (2024) find that an increase of one rank in income raises the perceived wealth rank by almost the same amount as an increase in one rank in wealth (0.169 and 0.195 respectively). Moreover, Figure 5 in the Online Appendix C shows that there is a strong association between misperceptions of income and wealth ranks.

The exclusion restriction requires that living in a district with many large firms in 2014 versus 2010 does not directly affect savings through a channel unrelated to the pay transparency policy. One may argue that potentially between 2010 and 2014, incomes and unemployment risks in large firms may have evolved differently than in small firms.¹⁰ Figure 7 and Figure 8 in the Appendix D scrutinize these arguments by plotting both district-level employment and household disposable income against the share of large

⁹This is particularly the case since the within-firm perspective and the disaggregation of wages by different job types narrows down reference groups to include rather similar individuals. At the same time, the exemption of managerial positions from pay transparency requirements may add to more positive perceptions of treated individuals’ relative economic position.

¹⁰It is also conceivable that there is a retirement saving channel, if large firms started offering more generous packages to employees after the reform. However, this is unlikely to be a major driver of change because firm-based retirement options are very limited compared to countries such as the US or Switzerland.

firms before and after the implementation of the pay-transparency reform. Both figures suggest a rather flat relationship between the outcomes and enterprise demography. Crucially, the slope changes only minimally after the implementation of the reform. Moreover, it is important to stress that the policy evaluation that studies the effect of the wage transparency legislation on wages shows that there were no effects on this margin.

4 Data

For the main analysis, we employ data from the first three waves of the Austrian Household Finance and Consumption Survey (HFCS). For our novel instrumental variable approach we additionally match data on firm size at the district level.

Main analysis Since 2010 the HFCS is an ongoing harmonized household survey conducted throughout the euro area as an initiative of the European Central Bank (ECB). The Austrian HFCS is a repeated cross-section that is representative of the Austrian population. Much like the Survey of Consumer Finances (SCF) in the US, the HFCS collects detailed information on the balance sheet of households. The HFCS aspires to follow the high data quality standards implemented in the SCF.¹¹ Reliance on CAPI interviews and extensive consistency checks contribute to the data quality. Furthermore, the data providers offer multiple imputations to correct for non-response behavior and complex survey weights. From a battery of questions on assets and liabilities of each household, we obtain net wealth as the sum of both real and financial assets minus all household debt. In addition, the HFCS provides information on household and individual demographics, labor market outcomes, income and consumption. We use this information to construct control variables¹² and measures of savings.

The income measure includes wages, salaries, self-employed and property income, as well as monetary social transfers (including pensions). Taxes and social security

¹¹In contrast to the SCF, the Austrian HFCS does not engage in oversampling of wealthy households. While this can affect wealth aggregates and inequality indicators (Disslbacher et al. 2023; Kennickell, Lindner, and Schürz 2022), the extent to which the quantiles of the wealth distribution are affected by the undercoverage of the very wealthy is more limited. For example, Andreasch and Lindner (2016) show that underreporting in the Austrian HFCS can lead to discrepancies with the Financial Accounts. However, their microsimulation exercise indicates that simulating upper deposit amounts not well captured by the survey data has only a minor impact on statistics such as the median. Moreover, efforts to address deficient survey coverage at the top of the wealth distribution in Austria, for instance through Pareto imputation techniques, generally leave the quantiles used in this paper unaffected (Heck, Kapeller, and Wildauer 2020).

¹²Table 7 in Online Appendix A provides descriptive statistics for key household level variables.

contributions are deducted. Whenever we do not explicitly control for household size (as in the regression analysis), we use equivalized income. To that end, we employ the modified OECD-scale. While the analysis refers to the household level, we include some individual characteristics of the main respondent. Across waves, the average respondent is likely to be male and slightly above 50 years in age.

To construct a savings variable, we combine data on household consumption expenditure on non-durable consumption (including utilities) and spending on rent, as well as alimony payments.¹³ The savings rate s for each household h follows from dividing the monthly net household disposable income Y minus expenditure flow E by monthly net household income:

$$s_h = \frac{Y_h - E_h}{Y_h} \times 100 \quad (3)$$

Table 1 provides descriptive statistics for savings and savings rates. Across both HFCS waves, the share of households who engage in saving according to our definition is relatively stable and ranges from 94% to 95%. Mean (median) savings rates across all households are one (two) percentage points higher in 2010 than they are in 2017. Means and medians are closely aligned, though the latter are marginally higher across waves. It is important to note that our measure of savings differs from the conventional macroeconomic savings rate because it omits expenditures on durable goods. In our framework, funds that are regularly used for durable purchases are counted as savings, leading to substantially higher savings rates than those typically reported in macro figures. We offer alternative measures of savings in our robustness checks that suggest that our finding of significant over-saving among underestimators holds. This includes a savings measure that captures active saving where respondents provide information on how much money they can put aside each month. This measure is significantly lower than the estimates in Table 1. However, for the most part of the analysis, we choose to construct saving from data on income and consumption, since active saving is only available from 2014 onward in the survey.

Capitalizing on the flexibility of household surveys and their capacity to elicit subjective information, the Austrian HFCS also collects a wide range of information on individual attitudes, preferences and perceptions. We draw on a special question, asking the main respondent in each interview to situate their household in the national net wealth distribution. The question reads as follows: “If you consider the entire net wealth

¹³In the Online Appendix B, we list all key variables related to savings and income, along with the exact wording of the corresponding survey questions (translated to English).

Table 1: Savings rates descriptive statistics

Statistic	Savers 2010	Population 2010	Savers 2014	Population 2014	Savers 2017	Population 2017
Mean saving	1281.7	1182.7	1321.1	1252.2	1439.1	1357.4
Median saving	969.8	897.6	1000	943.4	1126.2	1061.4
Mean savings rate	47	43	46	43	45	42
Median savings rate	50	47	47	45	47	45
Population share	93.8	100	95.4	100	95.1	100
Observations	2236	2380	2838	2997	2910	3072

Note: This table provides summary statistics for savings. Savings derive from the difference between net disposable household income and consumption spending (net of durables, loan repayments, insurance policies, renovation, etc.). Dividing by net monthly household income yields savings rates. Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

of your household, which position in the net wealth distribution do you think your household occupies?”.¹⁴ The respondents can then either name a decile rank, or choose the appropriate decile using a slider. Figure 1 plots a smoothed estimate of the difference between self-declared decile rank and decile rank based on the data on household wealth from the survey against the CDF of net wealth.¹⁵

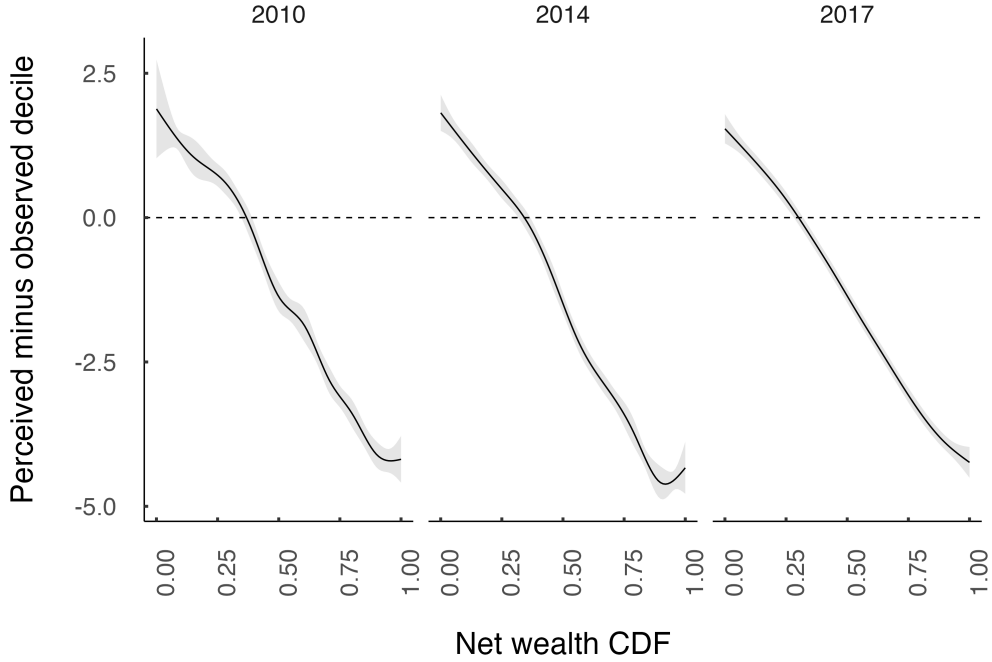
Positive values of the smoothed estimate of the bias imply that respondents overestimate their rank in the distribution, whereas negative values result from underestimation. The graph illustrates a negative association between decile rank and biased perception of rank in the wealth distribution. This pattern is robust across different survey waves. In 2017, on average, respondents underestimate their rank in the wealth distribution by 1.38 deciles. Individuals overestimating their rank in the distribution tend to be biased to a lesser degree than underestimators. At the bottom of the distribution, the bias amounts to two deciles in difference to the actual distributional rank. At the other end of the wealth distribution, the bias is twice that size in absolute terms. Between the first and the second tercile, approximately, the bias changes from positive to negative in each wave.

Two remarks about Figure 1 are important, because they have implications for the ability of our research design to identify the effects of overestimating one’s rank on savings. First, it should be noted that due to the dispersion of net wealth, the absolute differences in wealth in the region where people on average overestimate their wealth

¹⁴The question features towards the end of the interview. Throughout the interview, the concept of “net wealth” is used several times, such that respondents are familiar with the term. The survey also asks explicitly and directly about net wealth, priming respondents to think of their entire portfolio including real estate and life insurances, net of debt.

¹⁵The smoothed line is based on a generalized additive model with a penalized cubic regression spline.

Figure 1: Self-perceptions along the net wealth distribution



Note: The x-axis represents the CDF of the net wealth distribution. The y-axis plots a smoothed estimate of the mean difference between perceived decile and decile based on survey-elicited net wealth. Estimates are constructed from averaging across implicates. Standard errors are computed using Rubin's rule. Survey weights are taken into account. Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

rank or have accurate assessments are much smaller than at the top of the distribution (where individuals tend to be underestimators). Therefore, small errors in assessing one's own wealth relative to the wealth of others result in measured misperceptions at the bottom, whereas much larger assessment errors are required at the top. Second, overestimation in Figure 1 could arise from individuals who underreport certain types of assets, while holding accurate beliefs about their rank in the wealth distribution. In both cases, we would not necessarily observe differences between individuals who overestimate their wealth rank and those with accurate perceptions in terms of savings in the data.

Before proceeding with the analysis, we drop 70 observations without data on misperceptions in the 2010 survey, leaving us with a total of 8379 observations across three survey waves.

Instrumental variable approach To generate random variation in misperceptions, we capitalize on a policy promoting wage transparency in Austria. We use data from Statistics Austria on enterprise demography (from 2011 onward) to address the absence of workplace employee data in the HFCS. This data is available at the district level, and provides information on the prevalence of large firms, defined as those with at least 100 employees among all firms with any employees in 113 geographical units. Based on the regional identifiers in the HFCS, the enterprise demography data and the household survey data can be matched. This approach has two limitations: it cannot precisely measure firms just above the 150-employee threshold where the policy takes effect, and district-level firm size data for 2010 is unavailable. Although this imprecision likely introduces noise that attenuates the estimated relationship, the persistence of significant effects implies that the true causal impact may be even larger. We address this limitation in Online Appendix [D](#).

5 Results

This Section presents findings on biased perceptions and economic outcomes. Primarily, we focus on savings behavior as the main result and estimate predictive effects. Subsequently, this Section appraises the robustness of the findings, both in view of covariates and the operationalization of the dependent variable. In addition, our results feature a causal analysis based on the instrumental variable identification strategy.

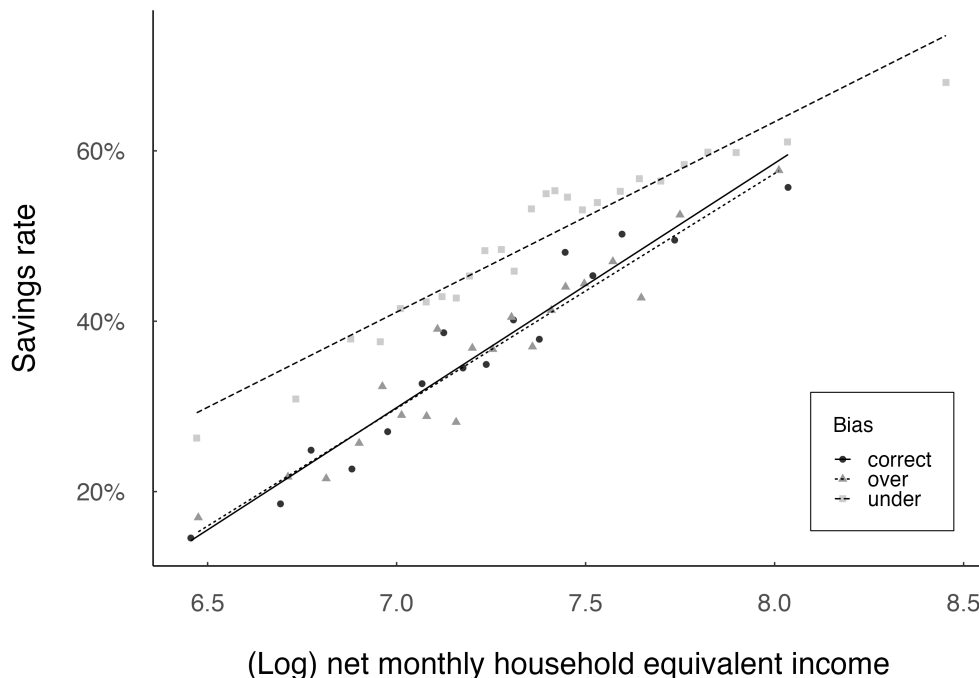
5.1 Main results

Figure [2](#) displays the main result of this paper at a glance. It shows the savings rate as a function of (log) equilvalized income. The graph plots this relationship for overestimators, underestimators and those who correctly assess their rank in the wealth distribution separately using a binned scatter plot. Figure [2](#) reveals a strong correlation between biased perceptions and the monthly savings rate. At all income levels, respondents who underestimate their rank in the wealth distribution save most. Overall, this difference amounts to more than eleven percentage points.¹⁶ Given mean savings rates between 42% and 43%, this is an economically significant difference. In relative terms, savings among underestimators are by more than 25% higher than among individuals who place themselves in the correct decile of the wealth distribution. Overestimators do not differ from individuals with correct assessments. This implies a heterogeneous relationship

¹⁶This can also be inferred from the first column in Table [20](#) in the Online Appendix [E](#)

between biased perceptions and savings: it is mostly underestimation rather than overestimation that drives the relationship between biased perceptions and savings.¹⁷

Figure 2: Average savings rate across the income distribution



Note: The x-axis refers to equivalized monthly household net income. The y-axis represents monthly savings as a fraction of household net income. The solid line plots the relationship between income and savings for individuals with correct assessments of their position in the wealth distribution. The dashed line refers to those who underestimate their household's position in the wealth distribution (underestimators). The dotted line refers to overestimators. The plot controls for wave fixed effects. Estimates are constructed from averaging across implicates. Survey weights are taken into account. Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

The next set of results introduces control variables and offers variants of Figure 2 with different residualizations. In the first panel of Figure 3, we condition on demographic variables. They include the gender of the respondent, along with a second degree polynomial of their age. Considering savings conditional on these variables does not change the substantive conclusions drawn from Figure 2. The second panel shows that savings rates residualized for employment outcomes (employment status of the main respondent,

¹⁷Given the almost linear relationship between wealth rank and misperceptions in Figure 1, this implies a non-linear relationship between savings and wealth rank. Figure 4 in the Online Appendix A illustrates this relationship. At the lower part of the wealth distribution, the savings rate is positive, before declining and rising again in the upper four quintiles of the distribution.

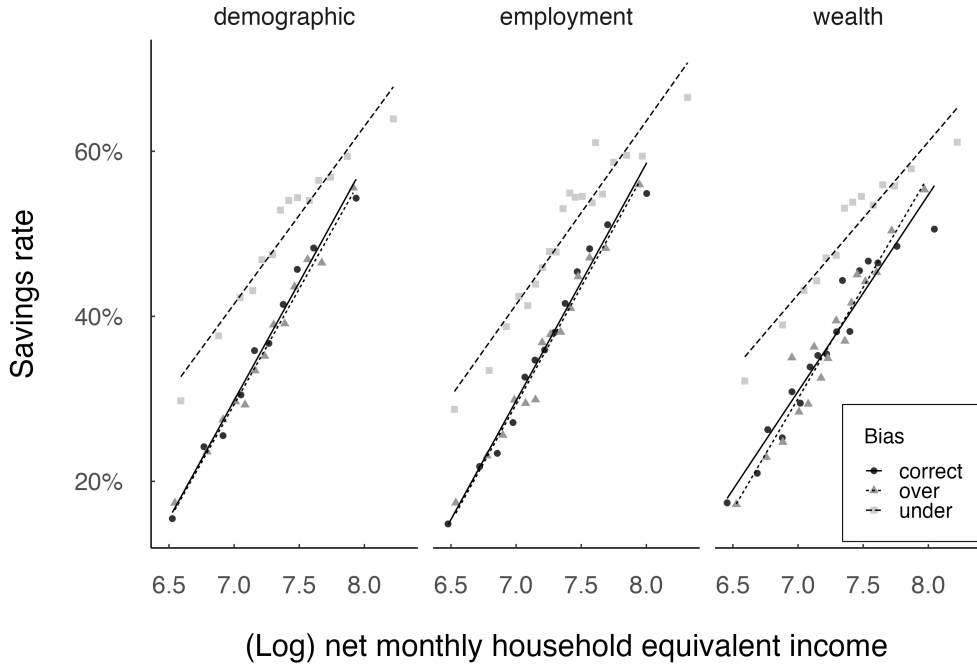
whether they work on a temporary contract, and four ISCO-based occupational indicators) do not change the conclusions either. The third panel in Figure 3 introduces wealth controls (ihs-transformed¹⁸ net wealth and a third-order polynomial of net wealth). This allows us to address the argument that both a downward bias in perceived rank in the wealth distribution as well as high savings rates are correlated with wealth, leading to a spurious correlation between biased perceptions and saving. Conditioning on wealth leads to a small change in the slope of the fitted line among all groups. However, even with the flexible controls for net wealth, the relationship between savings, income and misperceptions remains present in the data.

In addition to the graphical evidence, Table 2 provides regression results supporting our findings to establish the statistical significance of the results. In each column, the dependent variable is the savings rate in percent. Each specification pools data from the 2010, 2014 and 2017 waves of the HFCS, adding a wave fixed effect. We use the same measure of bias introduced in Figure 1, which exploits variation on the intensity of misperceptions. The first column estimates the difference in mean savings rates as a function of misperceptions (where positive values refer to overestimation), conditional on a respondent age polynomial, gender as well as household size and wave fixed effects. A one-decile increase in bias (for example, an individual in the fifth decile putting themselves in the sixth decile rather than the fifth) is associated with a 2.07 percentage points lower savings rate compared to individuals with accurate perceptions (no difference between actual and perceived wealth rank). The estimate is statistically significant at conventional levels. The wave fixed effects and the group indicator variables explain 18% of the variance in savings rates. Moving on to the next column, we introduce wealth and income controls in the form of ihs-transformed income and net wealth. The point estimate of the coefficient on biased perceptions increases to -1.11 percentage points, though it maintains statistical significance. The R^2 more than doubles, indicating a substantially improved fit. In the final column, we add a set of labor-market related individual controls to the specification. They include three educational dummies, seven industry dummies and controls for employment status. Compared to the specification in the second column, the changes in the coefficient on biased perceptions are marginal. The estimate suggests that each additional decile in underestimation results in a 1.36 percentage point increase in the savings rate.¹⁹ Crucially, the results rest on strong

¹⁸Inverse hyperbolic sine transformation

¹⁹Note, that Online Appendix E includes Table 20, which replicates Table 2 but using two indicator variables to distinguish over- and underestimators from individuals with accurate perceptions. We document that savings rates of underestimators exceed those of individuals with unbiased perceptions by five to eleven percentage points.

Figure 3: Residualized average savings rate across the income distribution



Note: The x-axis refers to equivalized monthly household net income. The y-axis represents monthly savings as a fraction of household net income. The solid line plots the relationship between income and savings for individuals with correct assessments of their position in the wealth distribution. The dashed line refers to those who underestimate their household's position in the wealth distribution (underestimators). The dotted line refers to overestimators. The graph controls for a second-degree polynomial of age, alongside the gender of the respondent in the first panel. The second facet is based on savings residualized for employment outcomes (ISCO job classification, temporary contracts and whether the respondent is employed). Savings in the third facet are residualized based on ihs-transformed net wealth, and third-degree polynomial of net wealth. All plots control for wave fixed effects. Estimates are constructed from averaging across implicates. Survey weights are taken into account.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

Table 2: Perceptions and savings: Main results

	Uncond diff (pp)	OLS I (pp)	OLS II (pp)	
Bias	−2.07*** (0.05)	−1.11*** (0.04)	−1.36*** (0.06)	
Net wealth (ihs)		0.22** (0.05)	0.14 (0.06)	
Net income (ihs)		24.15*** (0.31)	25.26*** (0.40)	[flushleft]
Wealth and income controls	No	Yes	Yes	
Employment controls	No	No	Yes	
Demographic controls	Yes	Yes	Yes	
Wave fixed effects	Yes	Yes	Yes	
R2	0.18	0.40	0.42	
Nobs	8379	8379	8379	

*Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size.*

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

extrapolation outside of the sparse common support of these groups. This is especially true at the tails of the distribution where under- (in the case of the first decile) and overestimation (in the case of the tenth decile) cannot exist logically.

As an extension of our baseline findings in Table 2, we estimate a regression model with interactions to examine whether the effect of misperceptions on savings varies systematically with net wealth. Detailed results of this analysis are presented in Table 23 in the Appendix. In addition to standard controls for wealth and income, the model incorporates interactions between the misperception indicators and net wealth (transformed via the inverse hyperbolic sine), thereby allowing us to derive predictive effects along the wealth distribution.

The results reveal that, from the 7th percentile onward, underestimators save substantially more than households that accurately assess their wealth rank. Specifically, at the 10th percentile, underestimators exhibit a savings rate approximately 2.3 percentage points higher than that of their accurately assessing peers, with this gap increasing to about 9.1 percentage points at the median and reaching roughly 12.5 percentage points at the upper end (P90). Overall, the predictive effect of underestimation increases by approximately 10 percentage points from the 10th to the 90th percentile, suggesting that underestimators save considerably more when they are in a position to “afford” higher savings relative to those who assess their decile correctly.

In contrast, overestimators display savings behavior that is statistically indistinguishable from that of the accurately assessing group across the wealth distribution. This pattern indicates that upward misperceptions do not significantly alter savings behavior, a finding that is not unexpected given that, in the lower part of the distribution, only a small nominal misperception is required to shift an individual into a higher decile. Moreover, some overestimators may end up being overestimators just because they underreport certain wealth components - possibly due to privacy concerns - which results in their perceived wealth rank aligning closely with their true position and their savings behavior therefore closely to those of households correctly assessing their wealth rank position.

These predictive effects substantiate that the mechanism by which misperceptions influence savings is predominantly driven by underestimation. The evidence thus suggests that subjective underestimation of wealth leads to economically meaningful increases in the savings rate, particularly as households become wealthier, while overestimation has no significant effect.

Our primary empirical findings, grounded in the minimal model presented in Section 2, are summarized as follows: Elevated savings rates among underestimators, as illustrated in Figures 2 and 3, lend support to Hypothesis (ii). For individuals underestimating their rank in the wealth distribution, it is consistent with the predictions of Equation 1. Moreover, the regression analysis presented in Table 2 reveals negative and significant coefficients on our measure of biased perceptions. This implies that negative bias (underestimation) is associated with higher savings, thereby providing further support for Hypothesis (ii). Concurrently, the positive coefficients associated with income and wealth variables reinforce Hypothesis (i), which suggests a positive relationship between savings rates and both income and wealth levels.

5.2 Sensitivity

The first step in appraising the sensitivity of our findings is the replication of the baseline results while controlling for wealth ranks and income as flexibly as possible. Table 3 summarizes this exercise. Each specification controls for personal and household demographics and employment-related characteristics of the respondent, ihs-transformed household net disposable income and wealth, as well as wave fixed effects (paralleling the specification in the baseline model in the third column in Table 2). The first column demonstrates that allowing for wealth rank dependent savings (in addition to the level of household net wealth) reduces the estimated coefficient compared to the baseline

specification (with all other control variables in the final column of Table 2) substantially. Underestimating one’s wealth rank by one decile is only associated with a 0.12 percentage point increase in the savings rate once we introduce nine indicator variables to flexibly control for a household’s wealth rank. Notably, wealth ranks do not explain the entire effect of misperceptions (which correlate with wealth ranks), and a statistically significant estimate of the correlation between misperceptions and savings remains. In addition, it is worth pointing out that flexibly controlling for wealth ranks does not significantly improve the model fit (compared to the third column in Table 2, R^2 increases by only three percentage points). The second column reveals that controlling for the CDF of net wealth also reduces the association between wealth rank misperceptions and savings rates. Once wealth rank in terms of the CDF is taken into account, the marginal increase in savings that is associated with an additional decile in underestimation reduces to 0.17 compared to 1.36 in the third column of Table 2. Even though the magnitude of the estimate is much lower than in the baseline model in Table 2, the estimate is still statistically significant at the five percent level. Finally, the third column in Table 3 reports the results controlling for an interaction between $\ln$ -transformed income and wealth. The substantive conclusion of Table 2 still holds, as the estimate of the association between bias in wealth rank perceptions and the savings rate is similar to the results reported in the baseline specifications both in terms of magnitude and statistical significance.

In a next step, we investigate whether the results are sensitive to the operationalization of saving in our main specification. We appraise whether the results hold among households who report that they put money aside every month, and whether the decision to put money aside actively is associated with perceptions.²⁰ While the dependent variable in the first column of Table 4 is the savings measure from Equation 1, the indicator variable in the second column and the savings rate in the third column are based on the amount of money that households actively put aside each month. Finally, the marginal propensity to save is the dependent variable the fourth column of Table 4. This variable is only available in the 2017 wave of the HFCS. It is constructed from a survey question where households allocate a hypothetical windfall income shock to consumption and savings (Drescher, Fessler, and Lindner 2020). The control variables are the same as in Table 3 the third column of Table 2.²¹

In the first column, we limit the sample to households who report that they put

²⁰As detailed in Section 4, this variable (AHI0420. For details on the question, see Online Appendix B.) is not available in the first wave of the HFCS.

²¹The differences in data availability across waves and the focus on active savers in the first column result in lower sample sizes in Table 3 compared to the other tables.

Table 3: Perceptions and savings: Flexible income and wealth controls

	Wealth deciles	Wealth CDF	Income x Wealth
Bias	-0.12* (0.06)	-0.17* (0.08)	-1.43*** (0.06)
Net wealth (ihs)	-0.18 (0.08)	-0.27** (0.05)	3.41* (0.97)
Net income (ihs)	23.15*** (0.37)	22.63*** (0.40)	29.77*** (1.11)
Wealth and income controls	Yes	Yes	Yes
Employment controls	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes
Wave fixed effects	Yes	Yes	Yes
R2	0.44	0.43	0.42
Nobs	8379	8379	8379

[flushleft]

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size. Wealth deciles refers to a set of 9 net wealth decile dummies, wealth CDF is a control variable referring to the probability of a household having less wealth than a given observation, and income x wealth is an interaction between household income and wealth (ihs transformed).

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

aside money each month. Both statistical and economic significance of the baseline results in Table 2 do not change substantially, although the absolute magnitude of the coefficient estimate drops somewhat compared to the third column of Table 2. Each additional decile of underestimation is associated with a 1.21 percentage point increase in the savings rate. The second column refers to a logit model. The outcome variable is an indicator distinguishing households who save from those with no monthly savings. The coefficient on a respondent's misperceptions of their wealth rank is negative and highly statistically significant at conventional levels. Note that the estimate has to be interpreted as log-odds. The third column of Table 4 tests for our finding's sensitivity to the measurement of the savings rate - the dependent variable. Instead of taking the ratio in Equation 1, the dependent variable in this column is the sum of debt repayments and self-declared monthly savings divided by disposable monthly net income. Compared to our baseline specification, the measurement of the savings rate influences our estimates. In our primary measure, savings are computed as the difference between income and

Table 4: Perceptions and savings: Additional results

	Savers (pp)	Participation (logit)	Active savings (pp)	MPS (pp)
Bias	-1.21*** (0.09)	-0.02*** (0.00)	-1.06*** (0.05)	-0.60* (0.22)
Net wealth (ihs)	0.15 (0.06)	-0.00 (0.00)	0.05 (0.05)	-0.21** (0.07)
Net income (ihs)	23.23*** (0.55)	0.20*** (0.01)	5.00*** (0.48)	1.10 (0.89)
Wealth and income controls	Yes	Yes	Yes	Yes
Employment controls	Yes	Yes	Yes	Yes
Demographic controls	Yes	Yes	Yes	Yes
Wave fixed effects	Yes	Yes	Yes	Yes
R2	0.39		0.11	0.02
Nobs	5119		6069	3072

[flushleft]

*Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size. The specification in the first column refers to the population with positive active savings only. Column 2 refers to a logit model with participation in active monthly saving as the dependent variable. Column 3 uses savings derived from a direct question on active monthly saving as the dependent variable. The dependent variable in the final column is the MPS.*

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

consumption, where consumption is defined to exclude durable goods expenditures. This means that expenditures on durables are not deducted from income - thereby increasing the calculated savings rate. In contrast, when using an alternative measure based solely on active savings (i.e., the amount households explicitly report setting aside each month), the estimated coefficient on misperceptions decreases in absolute magnitude, reflecting the generally lower level of active savings. Nonetheless, the relationship remains highly statistically significant, underscoring the robustness of our results. The last column of Table 4 documents a further drop in the magnitude of the coefficient measuring the association between misperceptions and savings, as savings are measured in terms of the MPS. However, the estimate is still significant at the 5% level and economically meaningful: For every decile of underestimation, an individual with a biased perception of their rank in the wealth distribution increases their MPS by 0.6 percentage points.

5.3 An IV approach to biased perceptions

The model specifications in the IV regression are most similar to the third column of the baseline results in Table 2. However, we add district fixed effects and reduce the sample to employed individuals, who are more likely to be affected by the policy

implementation than other groups such as retirees or the self-employed. Table 5 presents the results of the IV approach, dropping coefficients on control variables. The first column reports the coefficient where we instrument misperceptions. The coefficient implies that a one decile increase in the bias (where positive and negative values represent over- and underestimation respectively) leads to a 7.74 percentage point decrease in savings.

Column 2 shows the first stage with the magnitude of bias as the regressand. The interaction terms in Table 5 summarize the difference in the association between firm size and misperceptions between the first and the subsequent survey waves. Importantly, the instrument is statistically significant at conventional levels. The coefficient in the second row of Column 2 suggests that a one percentage-point difference in the share of large firms between two districts implies more than a third of a decile additional bias in 2014 compared to 2010. This is economically significant, since a one percentage point difference in the share of large firms in 2010 was associated with a difference of approximately half a decile in misperceptions. The F statistic reported at the bottom of the second column for the joint significance of the instruments suggests that the instruments are relevant.

Overall, the results in Table 5 indicate that the introduction of pay transparency is associated with variations in the prevalence of large firms at the district level, which in turn influence household misperceptions about their wealth rank and, consequently, their savings behavior. Given that the pay transparency reform serves solely as an informational intervention, its impact on savings should operate exclusively through changes in misperceptions. To capture this exogenous variation, we use as an instrument the interaction between time (pre- versus post-pay transparency) and the large firm share at the district level—a proxy for treatment intensity. The IV estimates reveal that, even after addressing endogeneity, the effect of biased perceptions on savings remains both large and statistically significant.

6 Summary remarks

The savings decision of the consumer is at the heart of macroeconomics. This paper sheds light on a vital yet often overlooked aspect of this decision, namely individuals' subjective perceptions.

Our findings illuminate a significant disparity between individuals' self-assessed wealth distribution ranks and objective reality, revealing a pervasive bias. More strikingly, this bias has substantial implications for economic behavior. Underestimating one's wealth

Table 5: Perceptions and savings: Instrumental Variables

	IV	First stage
Bias	-7.74* (3.45)	
Large firm share (2010)		44.91*** (11.90)
Large firm share X 2014		37.43*** (6.30)
Large firm share X 2017		11.53* (5.07)
Wealth and income controls	Yes	Yes
Employment controls	Yes	Yes
Demographic controls	Yes	Yes
Wave fixed effects	Yes	Yes
District fixed effects	Yes	Yes
F-Stat		28.17
Nobs	3798	3834
R2		0.45

[flushleft]

*Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size. IV refers to the coefficients from an instrumental variables regression, where perceptions are instrumented. The instrument is the interaction between survey wave and the share of firms with at least 100 employees by region (Large firm share). The corresponding first stage is labelled First stage. The dependent variable in the first column is the savings rate, while it is the magnitude of perception bias in the second.*

Source: HFCS 2010, 2014, 2017 - ECB and OeNB and Arbeitsstättenzählung 2011, 2014, 2017 (Registerzählung and Abgestimmte Erwerbsstatistik)- Statistics Austria

rank by 1 wealth decile goes along with a 2.07 percentage point higher savings rate, which reduces to 1.36 percentage points once a broad set of control variables are taken into account (predictive estimate). Evaluated at the average savings rate measure that we construct, 2.07 percentage points are equivalent to a 5% increase in the savings rate. Our IV estimate results in a substantial increase in the estimated magnitude of the effect: The (causal) IV estimate suggests that a one decile deviation in misperceptions towards underestimation increases the savings rate by 7.7 percentage points - equivalent to 18% of the mean savings rate. Our findings hold for various definitions of savings, and even extend in a weaker form to the marginal propensity to save.

Overall, our findings are consistent with most economic models of savings behavior which predict that savings increase in (permanent) income. However, the strong and statistically significant association between misperceptions of wealth ranks and savings suggests that an additional mechanism might be operative that is not part of the standard

framework. In our theoretical model, we show that preferences over relative wealth together with biased beliefs can reconcile our empirical findings with optimizing behavior.

While perceptions of relative affluence are certainly driven by demographics and individual personal traits, they are likely to evolve looking forward. Trends in residential segregation are likely to amplify the bias towards the middle. At the same time, the rise of social media and AI may provide an even stronger amplifier of biased beliefs. Online echo chambers allow individuals to select into groups of like-minded people with potentially similar economic backgrounds, giving homogeneous groups the opportunity to exchange views and perceptions. This behavior is facilitated by content algorithms matching individuals with information and other agents based on prior information about background, preferences and behavior.

Our contributions have far-reaching implications for policymakers. The existence of biased wealth perceptions opens avenues for targeted information treatments to influence macroeconomic trends and monetary policy transmission mechanisms. The determinants of savings rates and marginal propensities to consume are central to the transmission mechanism of monetary policy. Fiscal stimuli may also be more effective if paired with policies that aim at giving agents a better sense of their actual position in the wealth distribution.

Our analysis points to several avenues for future research. Most importantly, it would be interesting to study the implication of biased perceptions in view of other economic choices. This includes portfolio choice or labor market outcomes. In addition, our causal analysis could be supplemented by laboratory experiments to further explore heterogeneous treatment effects, for example. This would be particularly useful in view of designing policies that aim at alleviating bias in wealth rank perceptions.

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Online Appendix for
“Do biased perceptions of wealth distribution affect
savings behavior?”

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A Descriptive statistics

The HFCS provides core and derived variables. The derived variables include a measure of net wealth, where debt is subtracted from total household assets. Table 6 offers descriptive statistics for both net wealth and total assets for the 2010, 2014 and 2017 waves of the HFCS. In addition to means, the table provides the Gini index, net wealth at different cutoff points of the distribution, where q50 refers to the median. Moreover, table 6 summarizes quantile ratios, as well as the mean-to-median ratio.

Table 7 provides a compact overview of key variables featuring in the analysis.

Table 8 reports the mean savings rate for each decile of the equivalized net household disposable income distribution (by type of bias). Table 9 replicates this analysis, using median instead of mean savings rates.

A.1 Additional results

Table 10 presents OLS-regression results of the difference between perceived and computed rank in the wealth distribution. The dependent variable is the same in each model. The models differ simply in the regressors. The first specification "Resources" presents results for a combination of resource measures (wealth, income), leaving out some combinations that proved less relevant (such as higher order polynomials of income). It turns out that a third order polynomial of net wealth performs well in explaining differences in biased perceptions. All terms are statistically significant. In terms of the R^2 , the regression model in the first column can explain around 0.41 of the variation in biased perceptions.

Column 2 presents the estimates for a similar model, where we substitute the resource measures for control variables related to household and individual demographic characteristics and employment-related variables. A third-order age polynomial plays a key role as an explanatory factor. In addition, individuals born in Austria tend to be more prone to underestimate their relative position in the wealth distribution. Moreover, as household size increases, underestimation becomes a more prevalent phenomenon, while the opposite holds for an increasing number of dependent children. In view of employment characteristics, being employed in the first place is associated with an overestimation of one's household in the wealth distribution, a relationship which is significantly weaker for individuals in temporary contracts. All ISCO-dummies are statistically significant at conventional levels. The association between working in an ISCO 4 occupation and negative bias in perceptions is equivalent to increasing household size by one member. In addition to occupational characteristics, industry-level indicators also systematically

Table 6: Wealth descriptive statistics

Indicators	Net wealth	Assets
2010		
Mean	265032.6	281778.2
p90	542163	572580.8
p75	250469.8	275038.6
Median	76445	92792.4
p25	10314.8	12271.4
p10	977.4	2452.2
gini	76.2	73.4
Ratio p90/p10	554.7	233.5
Ratio p75/p25	24.3	22.4
Ratio p90/p50	7.1	6.2
Mean to median ratio	3.5	3
2014		
Mean	258413.7	275656.5
p90	518072.4	545515.8
p75	304572.5	334182.4
Median	85914.2	100431.6
p25	10642.4	12361
p10	994.2	2167.4
gini	73.1	70.9
Ratio p90/p10	521.1	251.7
Ratio p75/p25	28.6	27
Ratio p90/p50	6	5.4
Mean to median ratio	3	2.7
2017		
Mean	250272.5	269038.2
p90	524783.2	550209.6
p75	274657.4	311977.8
Median	82680.6	97185.2
p25	12735.4	14393.8
p10	2004	3200
gini	73	70.9
Ratio p90/p10	261.9	171.9
Ratio p75/p25	21.6	21.7
Ratio p90/p50	6.3	5.7
Mean to median ratio	3	2.8

Note: This table provides summary statistics for net wealth and assets.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

Table 7: Key variables descriptive statistics

Variable	Min	Median	Mean	Max
2010 (N= 2380)				
Size (count)	1	2	2.1	9
Male (1)	0	1	0.6	1
Age (years)	17	51	51.9	98
Net wealth (€)	-602.1	76.4	265	17398.1
Net income (€)	97.6	2000	2305.9	47500
2014 (N= 2997)				
Size (count)	1	2	2.1	8
Male (1)	0	1	0.6	1
Age (years)	18	54	54	85
Net wealth (€)	-504	85.9	258.4	43733.7
Net income (€)	498.4	2121.6	2449.9	18000
2017(N= 3072)				
Size (count)	1	2	2.1	8
Male (1)	0	1	0.6	1
Age (years)	17	54	53.8	85
Net wealth (€)	-636.6	82.7	250.3	42843.5
Net income (€)	300	2400	2718.2	100000

Note: This table provides summary statistics for key variables. Net wealth in thousands. Net income refers to monthly household net income. Size is household size. Male assumes unity for male reference person. Age refers to reference person. Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

Table 8: Mean savings rates for different perception biases and income deciles

Household income decile	Underestimator	No bias	Overestimator
1	27.2	17.6	18.6
2	34.1	28.2	27.8
3	39.8	30.2	28.7
4	44.8	33.1	33.7
5	49.4	37	34.7
6	52	45.1	46.4
7	52.2	42.9	40.9
8	52.7	45.6	42.9
9	56.5	50.5	50.4
10	58.6	53.1	55.7

Note: This table provides savings rates for different perception biases for the ten deciles of equivalized household net disposable income. Savings are the difference between net disposable household income minus the sum of spending on consumer goods, rent, and alimony payments. Dividing by net monthly household income yields savings rates. Source: HFCS 2017 - ECB and OeNB.

Table 9: Median savings rates for different perception biases and income deciles

Household income decile	Underestimator	No bias	Overestimator
1	28.7	12.9	17
2	35.6	28.9	28.2
3	41.7	29.8	27.7
4	47.1	30.4	33.9
5	49.8	37.1	33
6	54.8	48.5	47.1
7	55.6	45.6	43.6
8	54.6	48.2	45.9
9	58.7	53.7	53.1
10	61.9	57	58.6

Note: This table provides savings rates for different perception biases for the ten deciles of equivalized household net disposable income. Savings are the difference between net disposable household income minus the sum of spending on consumer goods, rent, and alimony payments. Dividing by net monthly household income yields savings rates. Source: HFCS 2017 - ECB and OeNB.

vary with the bias in perceptions.

The third column documents the explanatory purchase of subjective variables, including expectations. As trust increases by one unit on a Likert-scale from 1 to 10, positive bias increases by 0.06. The opposite is true for time preferences. People who care more about the future are those more prone to underestimation. Compared to the absolute magnitude of risk aversion, though, these effects are more limited. A one-point decrease in risk aversion on a 1 to 10 scale (1 being high risk aversion) is associated with an increase in the bias, as underestimation becomes more prevalent. Regarding income expectations, more optimistic individuals are more likely to underestimate their relative wealth position. Individuals who believe that they can rely on their family to help them out financially are more likely to overestimate their rank in the wealth distribution.

The fourth column combines all regressors from the previous three columns. By definition, R^2 is at its maximum in this column, reaching a level of 0.46. However, compared to the model in column 1, the additional variation explained by demographics, employment characteristics and behavioral traits is limited. In terms of statistical significance, demographic and behavioral variables maintain their relevance in large parts. At the same time, many of the occupational and industry dummies lose some of their explanatory power.

The variable selection in the last column is based on a best subsets regression for the 15 most powerful predictors among all predictors employed in experimenting with the other models in table 10. Therefore, some additional variables not represented in the previous columns feature here. In the spirit of the results in the fourth column, the best subsets regression suggests that resources, along with some demographic variables and behavioral traits are most important in explaining biased perceptions of one's rank in the distribution of wealth.

Overall, the results point towards a strong relationship between resources (income and wealth) and biased perceptions. However, this may partly result from reverse causality: Underestimating one's own wealth rank is associated with higher savings, which contribute to wealth accumulation in turn. Beyond income and wealth, demographic characteristics play a major role in shaping misperceptions - including household composition. Finally, behavioral variables such as trust have strong predictive power regarding perceptions of distributional ranking.

Tables 11, 12 and 13 illustrate the common support problem. Table 11 reports the number of respondents with correct assessments of their rank in the wealth distribution by decile of the net wealth distribution (rows) and decile of the distribution of household net disposable equivalized income (columns). Table 12 refers to underestimators, while

Table 10: Determinants of biased perceptions

	Resources	Demographic	Traits	All	Optimal
(Intercept)	-3.40** (0.88)	0.29 (0.18)	-2.17*** (0.12)	-5.28*** (0.85)	-3.07** (0.85)
poly(wlth_net, 3)1	-23.85* (11.77)			-22.22* (11.06)	-22.65* (11.10)
poly(wlth_net, 3)2	45.48*** (12.03)			41.04*** (10.79)	41.67*** (10.91)
poly(wlth_net, 3)3	-37.72* (15.48)			-34.24* (14.68)	-35.00* (14.70)
ihs(wlth_net)	0.21 (0.12)			0.18 (0.12)	0.20 (0.12)
ihs(ahi0710)	0.51** (0.12)			0.80*** (0.11)	0.83*** (0.11)
syear2017	0.19*** (0.04)	0.11* (0.05)		0.12** (0.04)	
ihs(wlth_net):ihs(ahi0710)	-0.05* (0.02)			-0.05* (0.02)	-0.05* (0.02)
poly(dhageh1, 3)1		-57.54*** (1.20)		-20.27*** (1.87)	
poly(dhageh1, 3)2		21.69*** (1.03)		16.76*** (1.27)	
poly(dhageh1, 3)3		5.13*** (1.21)		-2.78** (0.93)	
dhgenderh1		0.11* (0.04)		0.12*** (0.02)	
dem_birthAT		-0.82*** (0.06)		-0.30** (0.07)	-0.31** (0.07)
dem_hhsize		-0.76*** (0.03)		-0.44*** (0.04)	-0.45*** (0.03)
dem_kidsdep		0.53*** (0.03)		0.31*** (0.04)	0.31*** (0.03)
dem_partnered1		-0.27*** (0.04)		-0.01 (0.04)	
dem_single1		-0.37*** (0.04)		-0.18*** (0.03)	
emp_isco.2		-0.30*** (0.07)		0.21*** (0.05)	
emp_isco.3		-0.32** (0.08)		0.20* (0.08)	
emp_isco.4		-0.76*** (0.08)		0.07 (0.09)	
emp_isco.0		0.59*** (0.12)		0.22 (0.11)	
emp_nace.Agriculture..Forestry		-1.41** (0.40)		-0.64* (0.25)	
emp_nace.Mining..Manufacturing		-0.14* (0.05)		-0.10* (0.04)	
emp_nace.Construction		0.09 (0.10)		0.05 (0.06)	
emp_nace.Wholesale..Retail..Ac		0.21*** (0.04)		0.18*** (0.04)	
emp_nace.Transportation..Infor		0.46*** (0.08)		0.19* (0.08)	
emp_nace.Financial.and.Insuran		-0.58*** (0.08)		-0.16 (0.09)	
emp_nace.Real.Estate..Renting.		0.39*** (0.06)		0.40*** (0.06)	
emp_employ1		0.76*** (0.08)		0.14 (0.10)	0.13* (0.05)
emp_temp		-0.32** (0.11)		-0.35* (0.13)	
beh_trust			0.06*** (0.01)	0.05*** (0.01)	0.05*** (0.01)
beh_incexp			0.21*** (0.02)	0.18*** (0.02)	0.17*** (0.02)
beh_risklove			-0.21*** (0.02)	-0.20*** (0.01)	-0.17*** (0.02)
beh_timeP			-0.03*** (0.01)	0.01* (0.01)	
p_familysupport			0.69*** (0.04)	0.05 (0.03)	
dhageh1					-0.05*** (0.00)
dhageh1.3					0.00*** (0.00)
R2	0.41	0.20	0.02	0.46	0.45
Nobs	6048	6048	6048	6048	6048

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. The dependent variable is the magnitude of bias in real numbers

Source: HFCS 2014, 2017 - ECB and OeNB.

Table 13 provides counts for underestimators.

Table 11: Number of individuals with accurate self-perceptions by wealth and income deciles

Wealth decile/Income decile	1	2	3	4	5	6	7	8	9	10
1	150	55	34	25	15	8	4	3	4	4
2	50	58	36	24	19	16	8	4	2	4
3	21	31	38	41	44	35	40	18	12	8
4	10	15	21	18	27	23	44	25	16	7
5	7	9	12	20	12	20	18	29	33	22
6	2	5	3	5	13	10	12	15	19	18
7	0	3	3	2	9	9	7	8	8	13
8	0	0	0	0	2	1	2	1	3	8
9	0	0	0	0	0	0	0	0	2	3
10	0	0	0	0	0	0	1	0	2	3

Note: This table provides the number of observations belonging to the group of individuals who correctly assess their wealth rank in each decile of the household net wealth (rows) distribution by deciles of household net equivalized income (columns). The table refers to rounded averages across implicates.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

Table 12: Number of underestimators by wealth and income deciles

Wealth decile/Income decile	1	2	3	4	5	6	7	8	9	10
1	0	0	0	0	0	0	0	0	0	0
2	34	25	21	18	11	6	6	7	2	2
3	39	36	33	24	23	23	13	20	12	2
4	37	36	53	49	49	48	49	33	36	14
5	53	55	42	48	48	53	60	56	67	42
6	65	61	50	73	71	70	61	57	68	45
7	44	54	54	64	81	91	89	74	80	61
8	28	48	52	68	95	81	96	85	100	90
9	25	41	34	61	79	73	103	89	122	144
10	26	24	26	44	59	58	59	72	146	252

Note: This table provides the number of observations belonging to the group of individuals who underestimate their wealth rank in each decile of the household net wealth (rows) distribution by deciles of household net equivalized income (columns). The table refers to rounded averages across implicates.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

Figure 4 plots the savings rate according to Equation 1 against the CDF of net wealth. The smoothed line is based on a generalized additive model with a penalized cubic regression spline. In the lower part of the net wealth distribution, the savings rate falls approximately until the 10th percentile, before increasing steadily. This increase flattens out in the upper part of the distribution, exhibiting a kinked pattern. In 2010 and

Table 13: Number of overestimators by wealth and income deciles

Wealth decile/Income decile	1	2	3	4	5	6	7	8	9	10
1	131	137	73	67	46	34	35	24	27	8
2	59	88	83	99	69	49	42	29	19	6
3	24	35	38	70	50	59	62	42	26	16
4	8	13	16	25	28	45	55	50	47	24
5	1	7	8	13	10	25	20	22	38	23
6	1	3	1	2	6	5	4	6	12	17
7	1	0	0	0	1	2	2	1	5	5
8	0	0	0	1	0	0	0	0	0	0
9	0	0	0	1	0	0	0	0	1	0
10	0	0	0	0	0	0	0	0	0	0

Note: This table provides the number of observations belonging to the group of individuals who overestimate their wealth rank in each decile of the household net wealth (rows) distribution by deciles of household net equivalized income (columns). The table refers to rounded averages across implicates.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

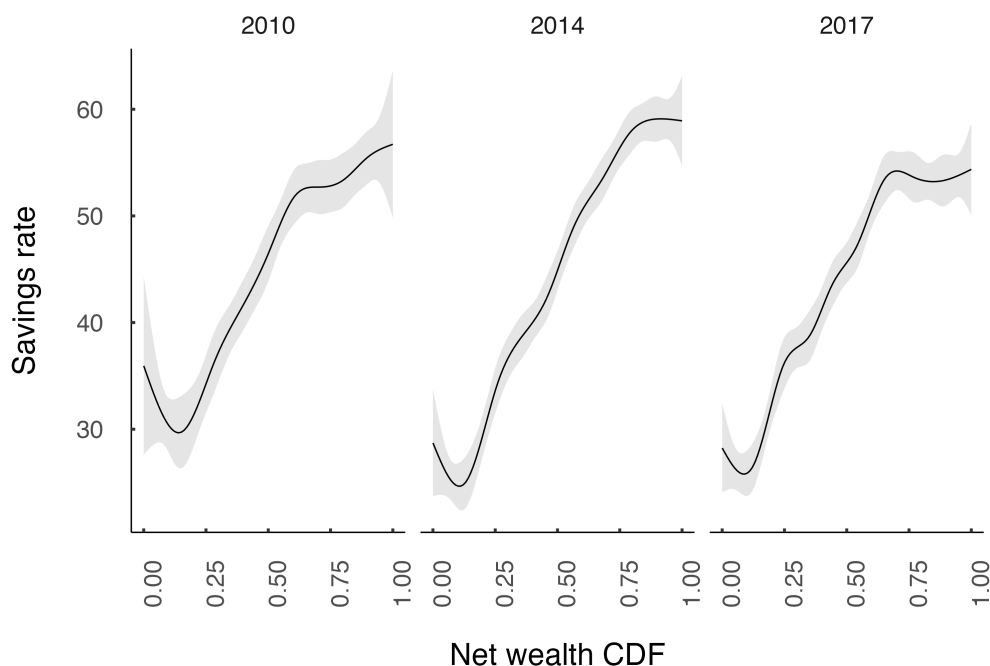
2017, the kink appears below the 75th percentile, whereas in 2014 it occurs approximately at the 75th percentile.

B Wording of survey questions

Our savings measure relies on several survey variables in the Austrian HFCS: AHNI0300, HB2300 as well as HI0310. These read as follows (translated to English):

- Total consumer expenditure (AHNI0300): *How much do you (or does your household) typically spend per month on all consumer goods and services? This includes expenses for food and beverages at home and away, utility costs, fees, leisure activities such as cinema or concerts, clothing, etc. Please exclude rent, taxes, financial payments (e.g., loan repayments, insurance premiums, pension contributions, etc.), and one-time payments (e.g., valuables, cars, large household appliances, furniture, etc.).*
- Rent (HB2300): *What is the monthly amount paid as rent (please exclude utilities, heating, etc. if possible)?*
- Alimony payments (HI0310): *About how much money do (you/your household) give per month?*

Figure 4: Wealth rank and savings rates



Note: This graph illustrates the correlation between the CDF of net wealth and savings rates. The smoothed line is based on a generalised additive model with a penalised cubic regression spline. Multiple imputations and survey weights are taken into account. Standard errors are computed using Rubin's rule.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

Alternatively, we measure savings as active saving and through the marginal propensity to save (MPS) in Table 4. The first measure is the sum of monthly savings (AHIO420) and debt repayment (DL2000 - derived variable referring to payments for a household's debt flow) - including payments into financial products that are earmarked for paying off a bullet loan in the future. The MPS refers to the HIZ040b question of the HFCS 2017. The key variables read as follows (translated to English):

- Savings amounts (AHIO420): Instructions: ASK *Do you usually have a certain amount of money left over each month that you can save or set aside — for example, for major purchases, emergencies, or building wealth?*. IF YES, THEN AHIO420: *How much?*
- MPS (HIZ040b): *Imagine you unexpectedly receive money from a lottery, equal to the amount of income your household receives in a month. What percent would you*

spend over the next 12 months on goods and services, as opposed to any amount you would save for later or use to repay loans? Instructions: READ IF PROMPTED: THE AMOUNT IS NET OF TAX) USE CARD, A RULER GOING FROM 0 TO 100, WITH THE ENDS LABELLED “I WOULD SAVE EVERYTHING AND/OR REPAY LOANS” AND “I WOULD SPEND EVERYTHING OVER THE NEXT 12 MONTHS” RESPECTIVELY. VARIABLE HIZ040b IS CALCULATED AS THE COMPLEMENT TO 100 OF THE ANSWER GIVEN

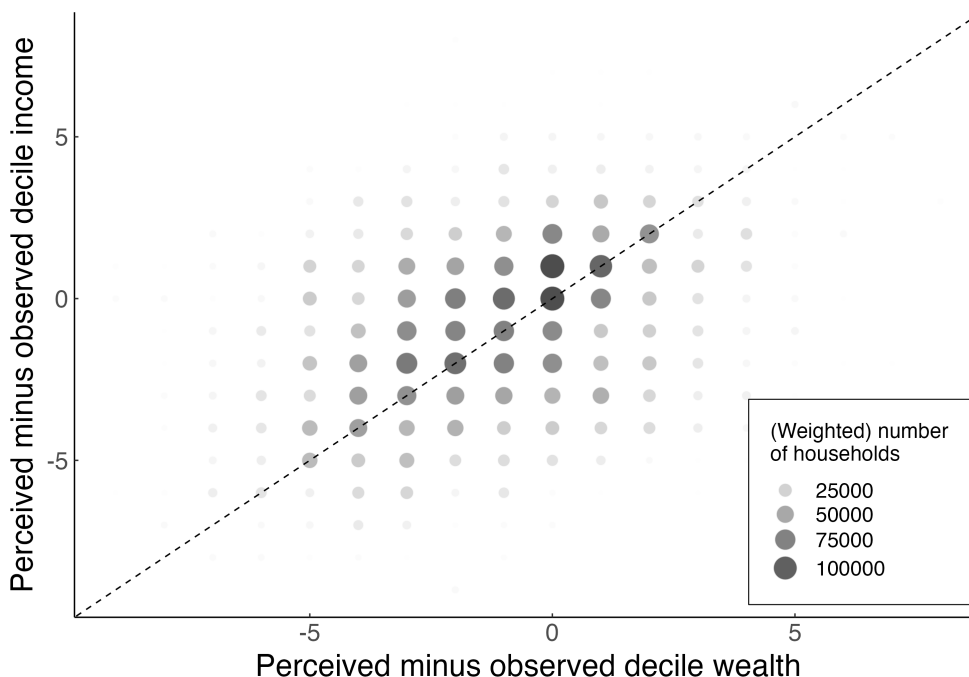
The savings rates (not MPCs) are calculated by dividing the saved amounts by disposable net household income (AHI0710). Income is elicited by posing the following question: *What would you estimate your (monthly disposable net income / your household’s monthly disposable net income) to be in total? I mean the total amount that comes from wages, salaries, income from self-employment, pensions, or retirement benefits. Please also include income from public benefits, income from renting or leasing, housing benefits, child benefits, and any other sources of income — and then subtract taxes and social security contributions. This refers to the total net income of all members of the household combined. Net income = income after taxes and social security contributions have been deducted.* Interviewers are provided with a list of income sources that should not be forgotten.

C Biased perceptions of wealth and income ranks

Data from the first wave of the HFCS (2010) allows comparing perceptions of relative wealth with respondents’ assessments of their position in the income distribution. Figure 5 explores the relationship between income and wealth rank perceptions. We grouped observations by the magnitude of bias in both income and wealth rank perceptions (ranging from -9 to +9). The shading and size of the bubbles refers to the weighted number of households in each group. The graph implies that a correlations exists between both types of bias. At the same time, biased perceptions of households’ relative ranks in the distribution of wealth are more pronounced.

It is conceivable that it is particularly wealthy individuals with high levels of debt who have biased perceptions of their rank in the distribution of net wealth. To investigate this claim, we consider misperceptions along the distribution of gross wealth. Figure 6 replicates the analysis in Figure 1 in Section 4, plotting misperceptions along the wealth distribution. However, instead of the CDF of the net wealth distribution, we use gross wealth in Figure 6. The results hold qualitatively. Individuals who may be asset rich

Figure 5: Misperceptions income versus wealth



Note: This graph illustrates the correlation of biased distributional rank perceptions in terms of income and wealth. Respondents are grouped by the magnitude of their bias in terms of net wealth and net monthly income rank. The size and color of the bubbles indicate the number of households in each group. Multiple imputations and survey weights are taken into account.

Source: HFCS 2010 - ECB and OeNB.

but indebted do not drive the conclusions that we draw on misperceptions.

C.1 Results with continuous bias

Here, we present the results of the main specifications, drawing on a different operationalization of biased perceptions. Rather than grouping individuals by the direction of bias into those who underestimate their position, as opposed to overestimators and respondents with accurate perceptions, we measure bias on a continuous scale in Tables 20 and 15.

Table 14: Perceptions and savings: Main results with discrete bias

	Uncond diff (pp)	OLS I (pp)	OLS II (pp)	
Perception bias	−1.09*** (0.03)	−0.77*** (0.04)	−0.80*** (0.04)	
Net wealth (ihs)		0.13*** (0.02)	0.15*** (0.02)	
Net eq. income (ihs)		6.27*** (0.28)	5.51*** (0.32)	[flushleft]
Wealth and income controls	No	Yes	Yes	
Personal controls	No	No	Yes	
Wave fixed effects	Yes	Yes	Yes	
R2	0.05	0.09	0.10	
Nobs	6048	6048	6048	

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Perception bias refers to perceived decile minus observed decile. Income and wealth controls refer to ihs-transformed household net wealth and monthly equivalent household net income. Personal controls include a second-degree age polynomial, three education dummies and seven industry dummies. pp refers to percentage points.

Source: HFCS 2014, 2017 - ECB and OeNB.

Table 15: Perceptions and savings: Interaction effects

	Uncond diff (pp)	OLS I (pp)	OLS II (pp)	
Perception bias	1.56*** (0.17)	1.00*** (0.15)	0.92*** (0.16)	
Perception bias X net wealth	−0.18*** (0.01)	−0.14*** (0.01)	−0.14*** (0.01)	
Net wealth (ihs)	0.53*** (0.03)	0.33*** (0.03)	0.35*** (0.03)	
Net eq. income (ihs)		5.56*** (0.23)	4.90*** (0.27)	[flushleft]
Income controls	No	Yes	Yes	
Personal controls	No	No	Yes	
Wave fixed effects	Yes	Yes	Yes	
R2	0.08	0.11	0.11	
Nobs	6048	6048	6048	

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Perception bias refers to perceived decile minus observed decile. Interaction terms are based on bias-dummies and ihs-transformed net wealth. Income controls refer to ihs-transformed monthly equivalent household net income. Personal controls include a second-degree age polynomial, three education dummies and seven industry dummies. pp refers to percentage points.

Source: HFCS 2014, 2017 - ECB and OeNB.

Table 16: Perceptions and savings: Main results with continuous bias and D2-D9 only

	Uncond diff (pp)	OLS I (pp)	OLS II (pp)	
Bias	−1.85*** (0.05)	−0.24** (0.07)	−0.33** (0.10)	
Net wealth (ihs)		3.02*** (0.14)	3.59*** (0.20)	
Net income (ihs)		22.86*** (0.29)	24.48*** (0.38)	[flushleft]
Wealth and income controls	No	Yes	Yes	
Employment controls	No	No	Yes	
Demographic controls	Yes	Yes	Yes	
Wave fixed effects	Yes	Yes	Yes	
R2	0.14	0.38	0.42	
Nobs	6722	6722	6722	

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

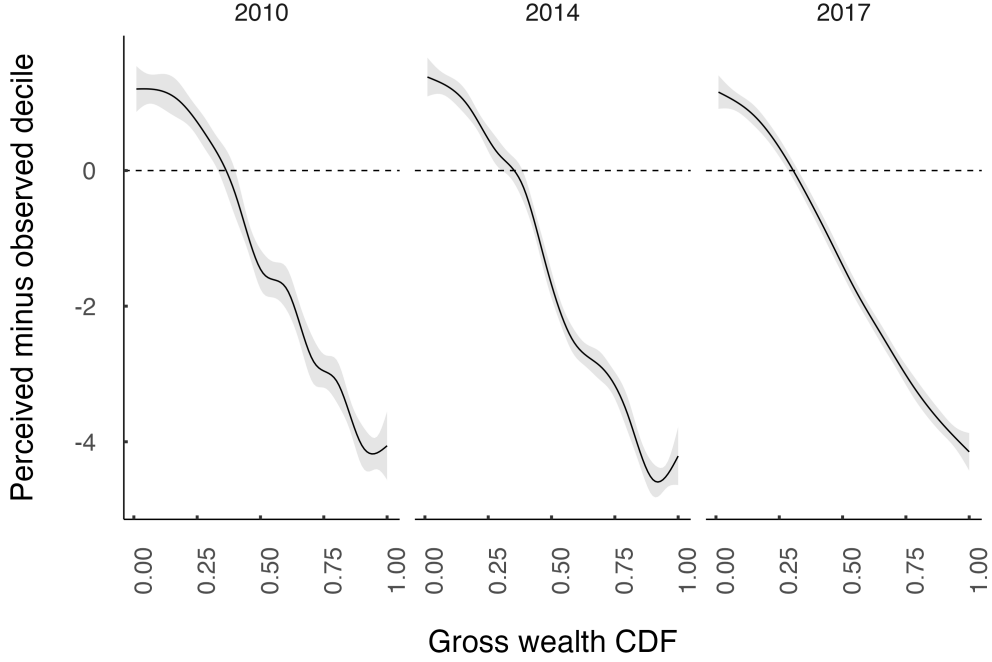
Table 17: Perceptions and savings: Interaction effects and D2-D9 only

	Uncond diff	OLS I	OLS II	
Perception bias	−0.21 (0.36)	−0.28 (0.36)	−0.29 (0.33)	
Perception bias X net wealth	0.08* (0.03)	0.08* (0.03)	0.07* (0.03)	
Net wealth (ihs)	3.28*** (0.09)	3.11*** (0.10)	3.22*** (0.12)	
Net eq. income (ihs)		1.41*** (0.24)	1.11*** (0.27)	[flushleft]
Wealth and income controls	No	Yes	Yes	
Personal controls	No	No	Yes	
Wave fixed effects	Yes	Yes	Yes	
R2	0.12	0.12	0.13	
Nobs	4848	4848	4848	

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Positive bias means overestimation, while negative bias is underestimation. Interaction terms are based on bias-dummies and ihs-transformed net wealth. Wealth and income controls refer to ihs-transformed household net wealth and monthly equivalent household net income. Personal controls include a second-degree age polynomial, three education dummies and seven industry dummies. Net wealth deciles 1 and 10 dropped.

Source: HFCS 2014, 2017 - ECB and OeNB.

Figure 6: Self-perceptions along the gross wealth distribution



Note: The x-axis represents the CDF of the gross wealth distribution. The y-axis plots a smoothed estimate of the mean difference between perceived decile and decile based on survey-elicited net wealth. Estimates are constructed from averaging across implicates. Survey weights are taken into account. Standard errors are computed using Rubin's rule.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

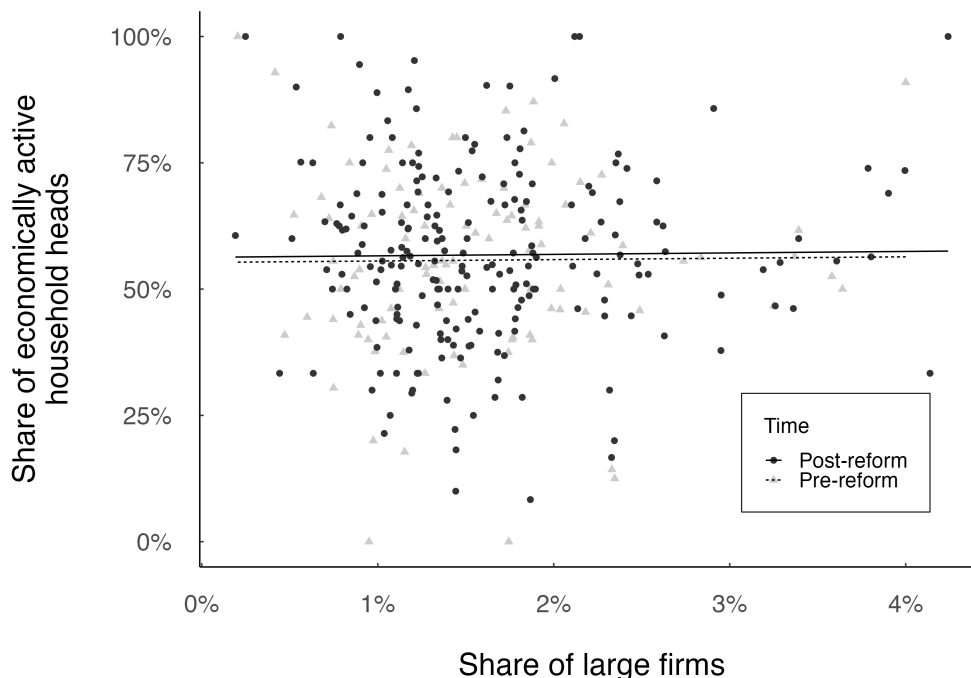
D Supplementary evidence on pay transparency and savings behavior

D.0.1 Districts by large firm prevalence over time

Figures 7 and 8 explore the possibility that the interaction of time with the share of large firms in a given district is related to employment or income. Households may save depending on their job-loss probability and their income. If there is a change in the relationship between the share of large firms in a district employment or income, that could lead to a direct relationship between the instrument and the savings rate - and thus a violation of the exclusion restriction. However, it turns out that the relationship between the share of large firms and the share of economically active household heads by district are flat. Crucially, this is the case before the implementation of the pay-

transparency law as well as after the reform.

Figure 7: Share of large firms and employment over time



Note: The x-axis represents the share of large firms in a district. The y-axis plots the average share of households headed by an economically active individual (not unemployed or retired). Pre-reform refers to the 2010 wave of the HFCS, post-reform refers to 2014 and 2017

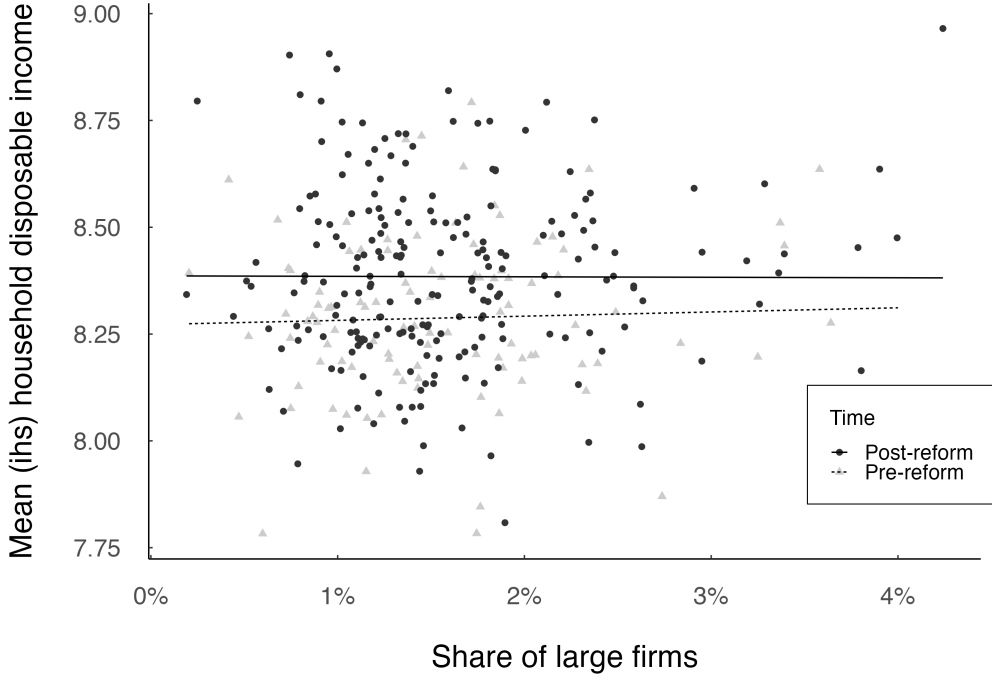
Source: HFCS 2010, 2014, 2017 - ECB and OeNB and Arbeitsstättenzählung 2011, 2014, 2017 (Registerzählung and Abgestimmte Erwerbsstatistik).

The relationship between average household disposable income in Figure 8 is also flat and relatively stable over time. While there is an indication that the slope of the curve is even lower after the reform, the change in slopes is insignificant.

D.0.2 Instrumental variables with leading values

Statistics Austria only provides data on enterprise demography from 2011 onward. To show that our instrument has no direct impact on savings and misperceptions prior to the reform, we merge the 2011 district-level data to the 2010 data from the HFCS. Due to the high persistence of the share of large firms in each district across time, we are confident that the 2011 data is a sufficiently good proxy for the missing 2010 information. Table 18 provides evidence for this argument. It replicates the results from Table 5, substituting

Figure 8: Share of large firms and household income over time



Note: The x-axis represents the share of large firms in a district. The y-axis plots the average household disposable income (inverse hyperbolic sine transformation). Pre-reform refers to the 2010 wave of the HFCS, post-reform refers to 2014 and 2017
Source: HFCS 2010, 2014, 2017 - ECB and OeNB and Arbeitsstättenzählung 2011, 2014, 2017 (Registerzählung and Abgestimmte Erwerbsstatistik).

not only the 2010 data for its leading values, but also using 2018 and 2015 data with the 2017 and 2014 HFCS waves respectively. The IV estimate in the first column does not change substantially (falling from -7.7 to -10.25 compared to the estimates of Table 5). The same holds for the first stage, that refers to the association between large firm prevalence and misperceptions before and after the reform. The coefficients maintain their signs, order of magnitude and statistical significance when compared to the results presented in Section 5.3.

D.1 Debt repayment

The measure of saving in the main specifications (including the Tables 2 and 23) includes debt repayment. Moreover, we also include contributions to savings vehicles earmarked for the repayment of debt repayable at the end of the maturity. In this section, we replicate the main results with a narrower definition of saving. Therefore, we only

Table 18: Perceptions and savings: Instrumental Variables

	IV	First stage	
Bias	-10.25*		
	(4.68)		
Large firm share X 2014		33.87***	
		(6.36)	
Large firm share X 2017		14.06*	
		(5.22)	
Wealth and income controls	Yes	Yes	[flushleft]
Employment controls	Yes	Yes	
Demographic controls	Yes	Yes	
Wave fixed effects	Yes	Yes	
District fixed effects	Yes	Yes	
F-Stat		21.88	
Nobs	3798	3834	
R2		0.45	

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size. IV refers to the coefficients from an instrumental variables regression, where perceptions are instrumented. The instrument is the interaction between survey wave and the share of firms with at least 100 employees by region (Large firm share). The corresponding first stage is labelled First stage. The dependent variable in the first column is the savings rate, while it is the magnitude of perception bias in the second.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB and Arbeitsstättenzählung 2011, 2015, 2018 (Registerzählung and Abgestimmte Erwerbsstatistik)- Statistics Austria

Table 19: Perceptions and savings: Without debt repayment

	Uncond diff	OLS I	OLS II	
Perceived below observed	3.59*** (0.13)	1.07*** (0.17)	1.02*** (0.19)	
Perceived above observed	-0.06 (0.14)	0.15 (0.16)	0.16 (0.14)	
Net wealth (ihs)		0.33*** (0.01)	0.31*** (0.01)	
Net eq. income (ihs)		5.02*** (0.23)	5.10*** (0.24)	[flushleft]
Wealth and income controls	No	Yes	Yes	
Personal controls	No	No	Yes	
Wave fixed effects	Yes	Yes	Yes	
R2	0.03	0.11	0.13	
Nobs	6048	6048	6048	

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Income and wealth controls refer to ihs-transformed household net wealth and monthly equivalent household net income. Personal controls include a second-degree age polynomial, three education dummies and seven industry dummies.

Source: HFCS 2014, 2017 - ECB and OeNB.

consider savings as measured by direct answers to the survey questions on monthly savings (Section 4 spells out the precise wording of the question). The narrower definition of savings leads to a lower savings rate overall. For example, in 2017, the share of savers in the population drops from 75% to 68.8%. At the same time, the mean savings rate for the full population decreases from 432.4 to 308. This corresponds to a drop in the savings rate by 3.8 percentage points. Among savers, the savings rate falls analogously.

Table 19 replicates the results from Table 2. The coefficients are smaller in absolute magnitude than those reported in the main specification. For example, the first column suggests that the savings rate among individuals who underestimate their rank in the wealth distribution exceeds the savings rate in the reference group by 3.59 percentage points. In the main specification, this coefficient is 6.23. Even though the coefficient magnitude falls, it is still precisely estimated and positive. This finding holds at conventional levels of statistical significance. The decline in the coefficient magnitude carries over to the second and the third column of the Table, introducing controls for resources (income and wealth) and further control variables respectively. Both coefficients on the indicator variable that identifies underestimators range slightly above one. While this implies a 3.5 percentage point decrease in the coefficient magnitude relative to the main specification, both coefficient estimates maintain their statistical significance. Much like in Table 2, the savings rates of survey respondents who overestimate their wealth rank

Table 20: Perceptions and savings: Main results with categorical bias

	Uncond diff (pp)	OLS I (pp)	OLS II (pp)
Perceived below observed	11.11*** (0.35)	5.26*** (0.35)	5.76*** (0.42)
Perceived above observed	−0.12 (0.24)	−0.69 (0.33)	−1.28 (0.54)
Net wealth (ihs)		0.22** (0.04)	0.18* (0.06)
Net income (ihs)		23.97*** (0.31)	25.04*** (0.41)
Wealth and income controls	No	Yes	Yes
Employment controls	No	No	Yes
Demographic controls	Yes	Yes	Yes
Wave fixed effects	Yes	Yes	Yes
R2	0.18	0.40	0.41
Nobs	8379	8379	8379

[flushleft]

*Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size.*

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

do not appear significantly different from savings rates among unbiased individuals.

E Additional regression specifications

The main specifications rely on a continuous measure of bias in perceived wealth ranks (the difference between perceived net wealth decile and actual net wealth decile). Alternatively, we group individuals into “underestimators”, “overestimators” and respondents with accurate perceptions. Next, we regress savings rates on this categorical variable, using the same control variables as in Table 2. The results mirror the findings from Figure 2. There is a large and significant coefficient in the first row of the table that suggests that the savings rates of individuals who underestimate their rank in the wealth distribution save by five to eleven percentage points more as a share of their income than individuals with accurate perceptions or “overestimators”.

Given a mean savings rate of 42% in 2017 (Table 1), this is an economically meaning-

Table 21: Perceptions and savings: Main results with continuous bias and D2-D9 only

	Uncond diff (pp)	OLS I (pp)	OLS II (pp)
Bias	-1.85*** (0.05)	-0.24** (0.07)	-0.33** (0.10)
Net wealth (ihs)		3.02*** (0.14)	3.59*** (0.20)
Net income (ihs)		22.86*** (0.29)	24.48*** (0.38)
Wealth and income controls	No	Yes	Yes
Employment controls	No	No	Yes
Demographic controls	Yes	Yes	Yes
Wave fixed effects	Yes	Yes	Yes
R2	0.14	0.38	0.42
Nobs	6722	6722	6722

[flushleft]

*Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size.*

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

ful effect (approximately 25% higher savings rates). All coefficients are highly significant in statistical terms.

Next, Table 21 limits the sample by dropping observations in the top and bottom decile of the net wealth distribution. By definition, individuals in these groups have more limited scope for misperceiving their wealth rank: Those at the top of the distribution cannot overestimate their rank, while respondents at the bottom cannot underestimate their rank. Therefore, one may want to appraise the robustness of our results to those that are not at either extreme of the distribution. Replicating the analysis presented in Table 2, Table 21 suggests that the sample restricting affects the magnitude of the coefficients, though they remain highly significant in statistical terms. For example, the coefficient on the magnitude of the bias in perceptions in the first column (only controlling for demographic factors) is -1.85, compared to -2.07 in the main specification.

Exploring the role of real estate wealth for misperceptions, Table 22 replicates the main analysis. However, the Table introduces controls for homeownership. The control variables affect the estimates substantially. For example, in the first specification,

Table 22: Perceptions and savings: Controlling for homeownership

	Uncond diff	OLS I	OLS II	
Bias	−0.08 (0.05)	0.31*** (0.04)	0.32*** (0.04)	
Homeowner	17.22*** (0.26)	14.22*** (0.25)	14.19*** (0.24)	
Net wealth (ihs)		0.03 (0.03)	0.05 (0.03)	[flushleft]
Wealth and income controls	No	Yes	Yes	
Employment controls	No	No	Yes	
Demographic controls	Yes	Yes	Yes	
Wave fixed effects	Yes	Yes	Yes	
R2	0.26	0.46	0.46	
Nobs	8379	8379	8379	

Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size.

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.

underestimators engage in excess savings to a much more limited extent than in the baseline specification in Table 2. This is no surprise for several reasons, and should not be interpreted as evidence against a strong independent effect of perceptions. Firstly, homeownership in Austria is essentially a proxy for belonging to the upper half of the wealth distribution (Lindner and Schürz 2019). Therefore, there is a high level of collinearity between homeownership and wealth perceptions. Secondly, even if homeownership may lead individuals to underestimate their rank in the wealth distribution, our IV results suggest that removing the confounding effect of homeownership still leaves us with a considerable causal effect of perceptions.

Finally, Table 23 presents the interaction effects discussed in Subsection 5.1

Table 23: Perceptions and savings: Interaction effects

	Uncond diff (pp)	OLS I (pp)	OLS II (pp)
Perceived below observed	−35.40*** (1.62)	−11.92*** (1.53)	−12.79*** (1.52)
Perceived above observed	6.58*** (0.67)	2.27*** (0.41)	2.33*** (0.41)
Perceived below observed X net wealth	3.33*** (0.14)	1.32*** (0.14)	1.40*** (0.14)
Perceived above observed X net wealth	−0.78*** (0.09)	−0.40** (0.08)	−0.39** (0.07)
Net wealth (ihs)	1.08*** (0.05)	0.41*** (0.04)	0.43*** (0.04)
Net eq. income (ihs)		22.83*** (0.31)	23.46*** (0.30)
Wealth and income controls	No	Yes	Yes
Employment controls	No	No	Yes
Demographic controls	Yes	Yes	Yes
Wave fixed effects	Yes	Yes	Yes
R2	0.24	0.41	0.41
Nobs	8379	8379	8379

[flushleft]

*Note: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$. Multiple imputations taken into account. 100 replicate weights. All specifications feature the savings rate in percent as the dependent variable. Wealth and income controls refer to ihs-transformed household net wealth and monthly household net income. Employment controls (respondent) include three education dummies and seven industry dummies and controls for employment status. Demographic controls include respondent and household characteristics: a second-degree age polynomial, gender and household size.*

Source: HFCS 2010, 2014, 2017 - ECB and OeNB.